CARLOS FARIA

Electrical and Computer Engineer

ABOUT ME

Electrical and Computer Engineer with a background in embedded systems, hardware design, computer vision and real-time signal processing. Experienced in prototyping Raspberry Pi–based cyber-physical systems and sensor integration. Skilled in designing control and communication pipelines and low-latency data handling. Academic trajectory combines applied computer vision with strong foundations in electronics, control systems, and industrial communication protocols, enabling the development of end-to-end intelligent hardware solutions. Passionate about engineering systems that bridge physical devices and intelligent computation for real-world applications.

TECHNICAL SKILLS

- *Extensive experience in data science workflows, object-oriented programming and process automation.*
- *Solid understanding of ML theory, with practical academic and personal experience in developing and tuning ML models.*
- *Programming Languages: Python, Rust, C, C++, CSS, HTML.*
- *ML & Data: Pytorch, Numpy, Pandas, Matplotlib, Scikit-learn, OpenCV, Hugging Face Transformers.*
- Skilled in database creation, structuring, and advanced querying and creation using PostgreSQL and maintenance of ETL processes.
- Experienced in Git version control for collaborative software development.
- Knowledge of the ROS framework.

EDUCATION

FEUP – M.Sc. Electrical & Computer Engineering

SEP 2022 – JUL 2025 / PORTO, PORTUGAL

- *Thesis: "AI-driven Media Quality Monitoring System" (real-time video QoE prediction via Transformer architectures).*
- *Relevant Coursework and Contents: Computer Vision; Machine Learning; OOP programming; Large Language Models, Perception and Mapping, Nonlinear Control Systems.*

Fundação Getúlio Vargas – MBA Finance

APR 2018 – NOV 2020 / RIO DE JANEIRO, BRAZIL

- *Financial Risk Modeling and Forecasting.*
- *Corporate Finance; Financial Derivatives; Risk Management.*

CEFET-RJ – B.Eng. Mechatronics & Automation Engineering

MAR 2011 – JUN 2016 / RIO DE JANEIRO, BRAZIL

- *Final Project: Temperature control through PID tuning and Pulse-width modulation.*
- *Core Modules: Linear Control Systems; Embedded Systems; Industrial Communication Protocols.*

EXPERIENCE

BISECT LDA – AI/ML/Computer Vision Thesis Intern

FEB 2025 – JUN 2025, MATOSINHOS

- *Developed a state-of-the-art Computer Vision Transformer-based deep learning model for predicting video Quality using Pytorch.*
- *Designed and implemented a real-time inference pipeline in Rust to ensure low-latency deployment within streaming environments.*

FEUP – Research Assistant

NOV 2024 – JAN 2025, PORTO

- *Managed OS provisioning and wireless networking setup for a Raspberry Pi cluster, ensuring reliable set up for Industry 4.0 Cyber-Physical Systems (CPS).*
- *Built custom 3D-printed enclosures to support the physical deployment of the embedded cluster.*

RSB Comunicação na Imagem – Embedded Systems Developer Intern

MAY 2024 – OCT 2024, PORTO

- *Prototyped sensor-driven advertising kiosks on Raspberry Pi 5; optimized real-time Python pipelines for interactive media.*

Freelance Embedded Systems Developer

JUN 2024 – JUL 2024, PORTO

- *Developed peer-to-peer high fidelity audio transmitting edge device for the Raspberry Pi.*

Stone, Eventials, OSX, OGX – BI, Data and Financial Analyst

2016 – 2021, BRAZIL

- *Developed Python dashboards for operational KPI monitoring and predictive analytics (queuing, churn models).*
- *Optimized telemarketing team scheduling.*
- *Automated financial reporting pipelines and KPI monitoring using Python, Power BI, Google Data Studio and Metabase.*
- *Automated controllership and back-office processes leveraging performance and cost reductions.*
- *DB querying automation using PostgreSQL and Google BigQuery.*

CERTIFICATIONS

- Scrum Fundamentals Certified (SFCTM).
- IELTS English Proficiency Test with CEFR Level C1.

CONTACT ME:

Phone: +351 927 989 860
E-mail: crls.thadeu@gmail.com
LinkedIn: https://www.linkedin.com/in/carlos-faria-33b51291/
Github: https://github.com/cthadeufaria
Location: Porto, Porto, Portugal.